

Lab-4 Ex4 - Vectors and Dot Product

Write a C program to ask the user for two vectors v and w . We can model each vector as 3 double-type values that are stored in a **3-element array**, e.g., $v[0]$, $v[1]$, $v[2]$ and $w[0]$, $w[1]$, $w[2]$ for vectors v and w respectively. As a reminder, to declare v as a 3-element array, we will write:

```
double v[3];
```

Calculate and print the *dot product* (scalar product) of v and w , in 2 decimal places.

Hint: Please find out the definition of dot product on your own, in case you have forgotten what dot product is!

User input is ***bold and italic*** in the following sample runs.

Sample run #1:

Enter vector v ($[0]$ $[1]$ $[2]$): ***-4 0 2.5***

Enter vector w ($[0]$ $[1]$ $[2]$): ***3 -9.876 5***

Dot product $v \cdot w = 0.50$

Sample run #2:

Enter vector v ($[0]$ $[1]$ $[2]$): ***-3.25 6.8 -2.6***

Enter vector w ($[0]$ $[1]$ $[2]$): ***-5.00 -0.1 17***

Dot product $v \cdot w = -28.63$